Tribhuwan University

Institute of Science and Technology

Bhairahawa Multiple Campus

An internship report on

"Full Stack Development"

At Sajilo Digital Pvt. Ltd

(In partial fulfilment of the requirement of Bachelor of Science in Computer Science & Information Technology (BSc. CSIT))

Submitted By:

Bal Gobind Chaudhary (30286 / 078)

Submitted To:

Department of Computer Science and Information Technology

Bhairahawa Multiple Campus

Bhairahawa, Nepal

November, 2025

Under the supervision of

Mr. Suraj Pandey

SUPERVISOR RECOMMENDATION

This is to certify that the internship report titled "Full Stack Development of Mount Glacier Trek Adventure Tourism Platform" has been prepared by Bal Gobind Chaudhary under my academic supervision. This report is submitted in partial fulfillment of the requirements for the Bachelor of Science in Computer Science and Information Technology degree.

As his college supervisor, I monitored Bal Gobind's progress throughout his internship at Sajilo Digital Pvt. Ltd. He effectively applied his comprehensive knowledge of both frontend and backend technologies to develop a sophisticated tourism web application. His report thoroughly documents the complete software development lifecycle, including system architecture, database design, API development, and frontend implementation.

The report is well-structured, technically detailed, and demonstrates a mature understanding of full stack development principles. It meets and exceeds the academic standards expected for the BSc.CSIT program.

……

Mr. Suraj Pandey

Supervisor, BSc.CSIT

Bhairahawa Multiple Campus

LETTER OF APPROVAL

This is to certify that the internship report "Full Stack Development of Mount Glacier Trek Adventure Tourism Platform" prepared by Bal Gobind Chaudhary is submitted in partial fulfillment of the requirements for the Bachelor of Science in Computer Science and Information Technology degree. We have reviewed this report and find it comprehensive in scope, rich in technical content, and professionally organized. It adequately represents the substantial work completed during the internship period and satisfactorily meets the academic criteria for the degree.

<table>
<tr><td colspan="2">…….
Lucky Prasad Joshi
Campus Chief
Bhairahawa Multiple Campus</td></tr>
<tr><td>……
Sunil Kumar Shah
BSc. CSIT Coordinator
Bhairahawa Multiple Campus</td><td>…….
Suraj Pandey
Supervisor
Bhairahawa Multiple Campus</td></tr>
<tr><td>……..
Internal Examiner</td><td>……
External Examiner
IOST: Tribhuwan University</td></tr>
</table>

ACKNOWLEDGEMENT

I extend my sincere gratitude to all those who supported and guided me throughout my internship journey and during the preparation of this report.

My deepest appreciation goes to Sajilo Digital Pvt. Ltd. for providing me with the invaluable opportunity to work as a Full Stack Developer Intern. I am profoundly grateful to my company mentor, Mr. Shailesh Mishra, for his exceptional guidance, technical insights, and constant encouragement. His mentorship was instrumental in helping me navigate the complexities of full stack development and successfully deliver the Mount Glacier Trek project

I extend my heartfelt thanks to my college supervisor, Mr. Suraj Pandey, for his academic guidance, constructive feedback, and support throughout the internship and report writing process. His supervision ensured my work maintained academic rigor while solving real-world problems.

I am also thankful to the entire technical team at Sajilo Digital for their collaborative spirit, knowledge sharing, and supportive work environment that greatly enriched my learning experience.

Finally, I express my eternal gratitude to my family and friends for their unwavering support, patience, and encouragement throughout my academic and professional journey.

……..

Your sincerely

Bal Gobind Chaudhary

Symbol no: 30286 / 078

BSc. CSIT, Bhairahawa Multiple Campus

ABSTRACT

This report documents the 10-week internship completed at Sajilo Digital Pvt. Ltd. as a Full Stack Developer, focusing on the design, development, and deployment of the Mount Glacier Trek adventure tourism website. The primary objective was to create a comprehensive online platform that facilitates trek bookings, provides detailed information about trekking packages, and manages customer interactions for a tourism business. The project encompassed both frontend and backend development, implementing a complete MERN stack (MongoDB, Express.js, React.js, Node.js) application. Frontend responsibilities included building responsive user interfaces with ReactJS, implementing dynamic routing, state management with Context API and Redux Toolkit, and creating interactive booking forms. Backend development involved designing RESTful APIs with Node.js and Express, implementing secure user authentication with JWT, designing MongoDB schemas for bookings, treks, and users, and integrating payment gateway functionality. The platform features include user registration and authentication, detailed trek listings with filtering, a multi-step booking system with real-time availability, secure payment processing, admin dashboard for content and booking management, and responsive design for mobile and desktop users. This internship provided comprehensive exposure to the full software development lifecycle, database design, API development, security considerations, and deployment of a production-ready web application. It significantly enhanced my skills in both frontend and backend technologies and provided practical experience in building scalable, secure web applications.

Keywords: *Full Stack Development, MERN Stack, ReactJS, Node.js, MongoDB, REST API, JWT Authentication, Payment Integration, Tourism Platform*

TABLE OF CONTENT

SUPERVISOR RECOMMENDATION ii

LETTER OF APPROVAL iii

ACKNOWLEDGEMENT iv

ABSTRACT v

LIST OF ABBRIVIATION viii

LIST OF FIGURES ix

LIST OF TABLES x

CHAPTER 1: INTRODUCTION 1

1.1. Introduction 1

1.2. Problem Statement 1

1.3. Objectives 2

1.4. Scope and Limitation 2

1.5. Report Organization 3

CHAPTER 2: ORGANIZATION DETAILS AND LITERATURE REVIEW 5

2.1. Introduction to Organization 5

2.2. Organizational Hierarchy 5

2.3. Working Domains of Organization 6

2.4. Description of the Intern Department / Unit 6

2.5. Literature Review / Related Study 7

CHAPTER 3: INTERNSHIP ACTIVITIES 9

3.1. Roles and Responsibilities 9

3.2. Weekly Log 10

3.3. Description of the Project (s) Involved During Internship 20

3.4. Tasks / Activities Performed 20

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES 29

4.1. Conclusion 29

4.2. Learning Outcomes 29

REFERENCES 31

APPENDICES 32

LIST OF ABBRIVIATION

API – Application Programming Interface

CI / CD – Continuous Integration and Continuous Delivery/Deployment

CRUD – Create Read Update Delete

CSS – Cascading Style Sheet

GPS – Global Positioning System

HTTP - Hypertext Transfer Protocol

HTTPS – Hypertext Transfer Protocol Secure

JSON – JavaScript Object Notation

JWT – JSON Web Token

MERN – MongoDB Express React Node.js

ODM – Object Data Modelling

OWASP – Open Web Application Security Project

REST - Representational State Transfer

SQL – Structure Query Language

UI – User Interface

UX - User Experience

LIST OF FIGURES

Figure 2.2:Organizational Hierarchy of Sajilo Digital 5

LIST OF TABLES

Table 2.4: Internship Duration 7

CHAPTER 1: INTRODUCTION

1.1. Introduction

The adventure tourism industry in Nepal has experienced significant digital transformation, with online platforms becoming essential for trekking companies to reach global audiences, manage bookings, and provide comprehensive information to potential clients. This internship report details my work as a Full Stack Developer at Sajilo Digital Pvt. Ltd., where I developed the Mount Glacier Trek website—a complete online solution for trekking package management and booking.

The project involved building a full-featured web application using the MERN stack (MongoDB, Express.js, React.js, Node.js) that serves both customers seeking trekking adventures and administrators managing the business operations. The platform was designed to handle the complete customer journey from browsing trek packages and checking availability to making secure bookings and payments.

This comprehensive development experience provided practical exposure to database design, REST API development, user authentication, payment gateway integration, and responsive frontend implementation—all essential components of modern full stack web development.

1.2. Problem Statement

Mount Glacier Trek, a growing adventure tourism company, faced several challenges in managing their operations and customer interactions:

- **Manual Booking Processes:** Reliance on phone/email for bookings led to errors, double bookings, and inefficient customer service.
- **Limited Online Presence:** Basic website provided only static information without interactive booking capabilities.
- **Inefficient Customer Management:** No centralized system to track customer inquiries, bookings, or payment status.
- **Poor Information Accessibility:** Potential clients couldn't easily browse available dates, compare packages, or check real-time availability.
- **Administrative Overhead:** Staff spent excessive time managing bookings manually rather than focusing on customer service and operations.

This project aimed to address these challenges by developing a comprehensive digital platform that automates booking processes, provides real-time information, and streamlines business operations.

1.3. Objectives

The primary objectives of this full stack development internship project were:

- To design and develop a complete tourism management system with customer-facing frontend and administrative backend.
- To implement secure user authentication and authorization for both customers and administrators.
- To create a dynamic booking system with real-time availability checking and confirmation.
- To integrate secure payment gateway for online transactions.
- To develop an admin dashboard for managing treks, bookings, users, and content.
- To implement responsive design ensuring optimal user experience across all devices.
- To design scalable database architecture using MongoDB to handle trek data, user information, and booking records.
- To build RESTful APIs with Node.js and Express for seamless frontend-backend communication.
- To ensure application security through input validation, authentication middleware, and secure payment processing

1.4. Scope and Limitation

Scopes:

- Full stack development using MERN (MongoDB, Express.js, React.js, Node.js) technology stack.
- Frontend development with ReactJS including state management, routing, and responsive design.
- Backend development with Node.js and Express including REST API design and implementation.
- Database design and implementation with MongoDB including schema design and optimization.
- User authentication and authorization using JWT (JSON Web Tokens).

- Payment gateway integration for secure online transactions.
- Admin dashboard development for content and booking management.
- Deployment of both frontend and backend to production environment

Limitations:

- Mobile application development was beyond the scope of this project.
- Advanced features like real-time chat support, GPS tracking, or social media integration were not implemented.
- The payment gateway was integrated in test mode; full production implementation requires additional compliance.
- Multi-language support and currency conversion were not included in the initial version.
- Advanced analytics and reporting features were limited to basic booking statistics

1.5. Report Organization

This report is structured into four chapters to provide a logical and comprehensive overview of the internship:

- **Chapter 1: Introduction**
  Presents an overview of the project, including the background of the internship, a clear problem statement, the defined objectives, the technical scope and limitations, and the overall organization of this report.
- **Chapter 2: Organization Details and Literature Review**
  Discusses the background of Sajilo Digital Pvt. Ltd., including its organizational hierarchy and core working domains. It also covers the description of the intern department and a review of the foundational technologies and existing literature relevant to modern frontend development.
- **Chapter 3: Internship Activities**
  Serves as the technical core of the report, provides a detailed account of my roles, weekly progress, project descriptions, and specific tasks undertaken.
- **Chapter 4: Conclusion and Learning Outcomes**
  Summarizes the project's achievements and reflects on the professional and technical skills acquired.

Additional sections, including References, and Appendices (containing screenshots, source code snippets), are included at the end of the report to provide supporting documentation and evidence of the work performed.

CHAPTER 2: ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1. Introduction to Organization

Sajilo Digital Pvt. Ltd. is a forward-thinking digital solutions agency based in Butwal, Nepal. The company specializes in providing end-to-end technology services, including custom software development, web and mobile application creation, digital marketing, and IT consultancy. Driven by a mission to empower businesses through innovative and efficient digital tools, Sajilo Digital emphasizes agility, quality, and client-centric development methodologies. The company's commitment to adopting modern tech stacks and fostering a culture of continuous learning made it an ideal environment for an immersive internship experience.

2.2. Organizational Hierarchy

The company employs a flat yet functional hierarchy designed to promote efficiency and collaboration:

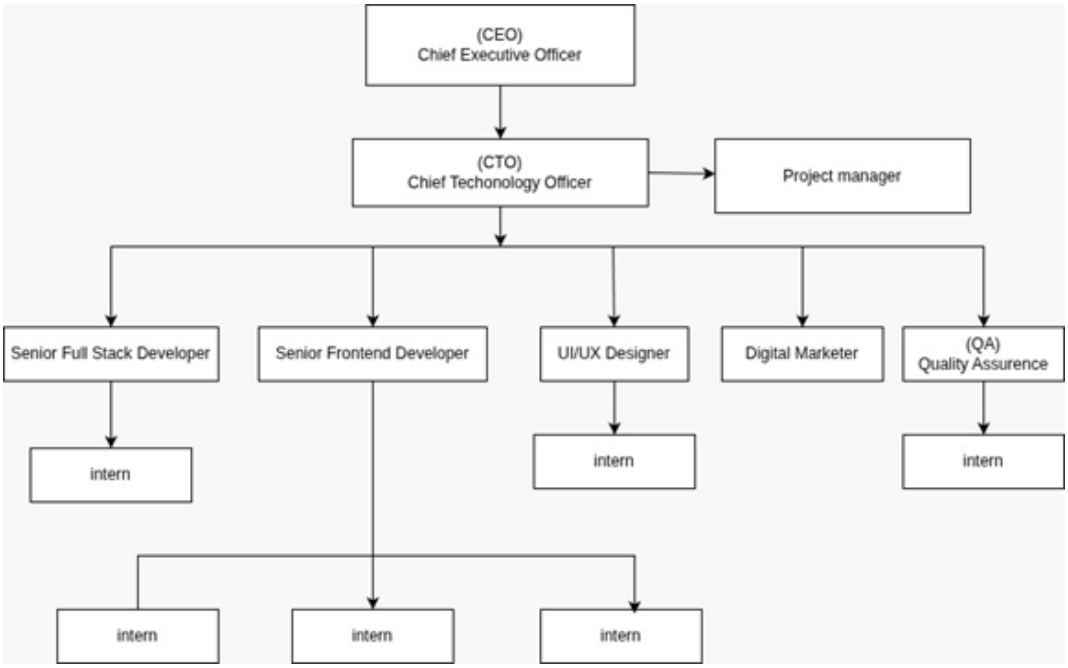

Figure 2.2: Organizational Hierarchy of Sajilo Digital

- Leadership / Directors: Set company vision, strategy, and oversee major client relationships.
- Project Management: Responsible for resource planning, timeline management, and acting as the bridge between clients and the technical team.

- Development Teams: Organized into frontend, backend, and full-stack units, tasked with designing, coding, and testing software products.
- UI/UX Design Team: Creates user interface mock-ups, prototypes, and maintains design systems.
- Quality Assurance (QA) Team: Ensures product reliability through systematic testing procedures.
- Marketing & Sales: Handles business development, branding, and client acquisition.
- Interns: Integrated directly into project teams to contribute to real-world projects under mentorship.

2.3. Working Domains of Organization

Sajilo Digital operates across several key domains within the tech industry:

- Custom Web Development: Building responsive websites, progressive web apps (PWAs), and single-page applications (SPAs) using stacks like Next.js, React, and Node.js.
- Mobile App Development: Creating cross-platform (React Native, Flutter) and native mobile applications.
- Enterprise Software Solutions: Developing tailored business management systems, CRMs, and ERPs.
- E-commerce Solutions: Designing and developing online stores with integrated payment and inventory systems.
- Digital Marketing & SEO: Enhancing online visibility and user engagement through targeted strategies.
- Cloud Services & DevOps: Assisting with deployment, hosting, and infrastructure management.

2.4. Description of the Intern Department / Unit

The internship program at Sajilo Digital is structured to provide maximum practical exposure. As a Full Stack Developer Intern, I was integrated into the web development team working on client projects. The department follows agile development practices with daily stand-ups, sprint planning, and code reviews.Interns participate in the complete software development lifecycle—from requirement analysis and database design to

implementation, testing, and deployment. This holistic approach provides comprehensive exposure to industry practices and collaborative development workflows.

Table 2.4: Internship Duration

Start Date:	11/17/2025
Daily Hours:	10:00 AM to 6:00 PM
Working Days:	Sunday to Friday
Position:	Full Stack Developer (MERN Stack)
Company Mentor:	Mr. Sailesh Mishra
College Supervisor:	Mr. Suraj Pandey

2.5. Literature Review / Related Study

The development of the Mount Glacier Trek platform was informed by contemporary full stack development practices:

MERN Stack Architecture: Resources like "Full-Stack React Projects" by Shama Hoque (2020) guided the implementation of the complete stack, emphasizing separation of concerns between frontend and backend layers.

REST API Design: Best practices from "REST API Design Rulebook" by Mark Masse (2011) informed the design of clean, predictable, and well-documented APIs for the booking system.

Database Design: Principles from "MongoDB: The Definitive Guide" by Kristina Chodorow (2013) guided schema design decisions, particularly for managing related data like treks, bookings, and users.

Authentication and Security: OWASP security guidelines and resources on JWT implementation informed the secure authentication system protecting user data and payment information.

Payment Integration: Documentation from payment gateway providers and security standards like PCI DSS informed the secure implementation of payment processing.

Responsive Design: Modern CSS frameworks and responsive design principles ensured the platform worked seamlessly across all devices.

CHAPTER 3: INTERNSHIP ACTIVITIES

3.1. Roles and Responsibilities

As a Full Stack Developer Intern, my responsibilities spanned both frontend and backend development:

Frontend Development (ReactJS):

- Developing reusable React components for trek listings, booking forms, and user interfaces.
- Implementing client-side routing with React Router for seamless navigation.
- Managing application state using Redux Toolkit for complex data flows.
- Creating responsive layouts using CSS Grid, Flexbox, and media queries.
- Implementing form validation and user feedback mechanisms.
- Integrating with backend APIs using Axios for data fetching.
- Optimizing frontend performance through code splitting and lazy loading.

Backend Development (Node.js/Express):

- Designing and implementing RESTful APIs for trek management, booking, and user authentication.
- Developing middleware for authentication, authorization, and input validation.
- Designing MongoDB schemas and implementing database operations.
- Implementing secure user authentication with JWT (JSON Web Tokens).
- Integrating payment gateway APIs for secure transaction processing.
- Implementing file upload functionality for trek images.
- Creating admin APIs for content management and reporting.
- Implementing error handling and logging mechanisms.

Database Development (MongoDB):

- Designing database schemas for users, treks, bookings, and payments.
- Implementing data relationships and referential integrity.
- Creating database indexes for performance optimization.
- Implementing data aggregation for reporting features.
- Ensuring data security through proper access controls.

3.2. Weekly Log

A detailed account of weekly activities, responsibilities, and learnings:

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 1	**Date: 11/17/2025 - 11/23/2025**

Responsibilities:

- Project initiation and requirement analysis.

Activities:

- Studied project requirements and business logic.
- Researched tourism platform features.
- Set up development environment and project repositories. Created initial project structure for both frontend and backend.

Observation:

- Understanding business domain thoroughly is crucial for designing appropriate technical solutions.

Performance Evaluation by Mentor:

- Demonstrated strong analytical skills in understanding business requirements and setting up a solid full-stack project foundation.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 2	**Date: 11/24/2025 - 11/30/2025**

Responsibilities:

- Database design and backend setup.

Activities:

- Designed MongoDB schemas for users, treks, bookings.
- Set up Express server with basic routing.
- Implemented initial API endpoints. Configured database connection and environment variables.

Observation:

- Well-designed database schema forms the foundation of a scalable application.

Performance Evaluation by Mentor:

- Showed good backend development skills by designing scalable database schemas and initializing API structure.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 3	**Date: 12/01/2025 - 12/07/2025**

Responsibilities:

- User authentication system.

Activities:

- Implemented user registration and login APIs.
- Added JWT token generation and verification.
- Created password encryption with bcrypt.
- Built authentication middleware for protected routes.

Observation:

- Security should be implemented from the beginning, not as an afterthought.

Performance Evaluation by Mentor:

- Implemented secure authentication mechanisms effectively with proper use of JWT and encryption practices.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 4	**Date: 12/08/2025 - 12/14/2025**

Responsibilities:

- Frontend foundation and authentication UI.

Activities:

- Set up React project with Redux Toolkit. Created authentication pages (login, register). Implemented protected routes and token management. Built responsive navigation and layout components.

Observation:

- State management architecture significantly impacts application maintainability.

Performance Evaluation by Mentor:

- Established a well-structured frontend with efficient state management and authentication flow.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 5	**Date: 12/15/2025 - 12/21/2025**

Responsibilities:

- Trek management features.

Activities:

- Developed backend APIs for trek CRUD operations. Implemented frontend pages for trek listing and details. Added image upload functionality for trek photos. Created filtering and search features.

Observation:

- File handling requires careful consideration of storage, security, and performance.

Performance Evaluation by Mentor:

- Successfully developed core application features with robust API integration and dynamic UI components.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 6	**Date: 12/22/2025 - 12/28/2025**

Responsibilities:

- Booking system development.

Activities:

- Implemented booking APIs with availability checking. Developed multi-step booking form in React. Added real-time date validation and participant limits. Created booking confirmation and email notification.

Observation:

- Complex business logic requires thorough testing and validation.

Performance Evaluation by Mentor:

- Demonstrated strong problem-solving ability by implementing complex booking logic with proper validation.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 7	**Date: 12/29/2025 - 01/04/2026**

Responsibilities:

- Payment gateway integration.

Activities:

- Integrated payment gateway APIs. Implemented secure payment processing flow. Added payment status tracking and webhook handling. Developed payment success/failure pages.

Observation:

- Financial transactions demand highest security standards and error handling.

Performance Evaluation by Mentor:

- Applied secure and reliable payment integration with proper handling of transaction flows.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 8	**Date: 01/05/2026 - 01/11/2026**

Responsibilities:

- Admin dashboard development.

Activities:

- Created admin APIs for managing bookings and users. Built admin dashboard with analytics. Implemented role-based access control. Added content management features.

Observation:

- Administrative interfaces require careful consideration of usability and data presentation.

Performance Evaluation by Mentor:

- Built a functional admin dashboard with effective role-based access control and data management features.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 9	**Date: 01/12/2026 - 01/18/2026**

Responsibilities:

- Testing, optimization, and security.

Activities:

- Conducted comprehensive API testing with Postman. Performed frontend testing and debugging. Optimized database queries and API responses. Implemented security enhancements and input validation.

Observation:

- Performance optimization often reveals architectural improvements.

Performance Evaluation by Mentor:

- Showed attention to performance and security through thorough testing, optimization, and validation.

……

Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bal Gobind Chaudhary

Project: Mount Glacier Trek Tourism Platform

Week no: 10	**Date: 01/19/2026 - 01/23/2026**

Responsibilities:

- Deployment, documentation, and handover.

Activities:

- Deployed backend to cloud platform. Deployed frontend to Vercel. Configured domain and SSL certificates. Created comprehensive project documentation and user guides.

Observation:

- Proper deployment and documentation ensure project sustainability and knowledge transfer.

Performance Evaluation by Mentor:

- Successfully handled deployment, configuration, and documentation, ensuring a stable and maintainable production system.

……

Mentor's Approval

Mr. Shailesh Mishra

3.3. Description of the Project (s) Involved During Internship

Primary Project: Mount Glacier Trek Tourism Platform

A complete booking and management system for adventure tourism services built with MERN stack.

- **Live URL:** https://www.mountglaciertrek.com.np
- **Technical Implementation:**
  i. Frontend: ReactJS with Redux Toolkit for state management, React Router for navigation
  ii. Backend: Node.js with Express.js framework, REST API architecture
  iii. Database: MongoDB Atlas (cloud) with Mongoose ODM
  iv. Authentication: JWT (JSON Web Tokens) with refresh token mechanism
  v. Payment: Integrated payment gateway for secure transactions
  vi. Hosting: Backend on cloud platform, Frontend on Vercel.
- **Key Features and Implementation:**
  vii. User Authentication: Secure registration, login, profile management
  viii. Trek Management: Detailed trek listings with images, descriptions, itineraries
  ix. Booking System: Multi-step booking with real-time availability checking
  x. Payment Processing: Secure online payment with multiple method support
  xi. Admin Dashboard: Comprehensive management interface for treks, bookings, users
  xii. Responsive Design: Mobile-friendly interface for all device types
  xiii. Email Notifications: Automated confirmation and reminder emails
  xiv. Search and Filter: Advanced trek search with multiple criteria.

3.4. Tasks / Activities Performed

This section provides a detailed account of the full stack development activities carried out during the 10-week internship for the Mount Glacier Trek tourism platform. As a Full Stack Developer, I was responsible for both frontend and backend development, database design, API implementation, payment integration, and deployment. Each activity is explained with context, approach, challenges encountered, and solutions implemented.

3.4.1 Project Planning and Requirement Analysis

The internship began with a thorough analysis of the tourism business requirements and technical planning.

Stakeholder Meetings: I met with the client (Mount Glacier Trek) to understand their workflow, booking processes, and pain points. We documented requirements for customer-facing features (trek browsing, booking, payment) and administrative features (trek management, booking overview, reporting).

Technical Stack Selection: Based on the requirements, I selected the MERN stack (MongoDB, Express.js, React.js, Node.js) for its scalability, developer productivity, and strong ecosystem. MongoDB was chosen for its flexible schema to accommodate varying trek details and booking data.

System Architecture Design: I designed a high-level architecture: React frontend communicating with a RESTful API built with Node.js/Express, data stored in MongoDB Atlas, authentication using JWT, and payment processing via a third-party gateway.

Project Setup: I initialized the repository with separate folders for frontend (client/) and backend (server/). I created a README.md with setup instructions and environment variables required.

3.4.2 Database Design and Implementation

A well-structured database was critical for the booking system.

Schema Design: I designed MongoDB schemas using Mongoose:

- **User:** fields for name, email, password (hashed), role (customer/admin), profile details.
- **Trek:** fields for name, description, itinerary, duration, price, difficulty, maxGroupSize, availableDates, images.
- **Booking:** fields for user reference, trek reference, date, participants, totalPrice, status (pending/confirmed/cancelled), paymentStatus, paymentId.
- **Payment:** fields for booking reference, amount, method, transactionId, status.

Indexing and Performance: I created indexes on frequently queried fields (userId, trekId, status) to optimize read performance. I also added compound indexes for booking date queries.

Data Relationships: I used Mongoose population to reference related documents (e.g., populating user details in bookings).

Data Seeding: I wrote seed scripts to populate initial trek data and a test admin user for development.

3.4.3 Backend API Development

I built a comprehensive RESTful API using Node.js and Express.

API Structure: I organized routes by resource (/api/auth, /api/users, /api/treks, /api/bookings, /api/payments). Each route had its own controller and service layer.

Authentication Endpoints:

- **POST /api/auth/register:** Creates a new user, hashes password with bcrypt, returns JWT.
- **POST /api/auth/login:** Verifies credentials, returns JWT.

User Endpoints:

- **GET /api/users/profile:** Returns user profile (protected).
- **PUT /api/users/profile:** Updates user details.

Trek Endpoints:

- **GET /api/treks:** Returns list of treks with pagination and filtering (by difficulty, duration, price range).
- **GET /api/treks/:id:** Returns single trek details.
- **Admin only:** POST, PUT, DELETE for trek management.

Booking Endpoints:

- **POST /api/bookings:** Creates a new booking after checking availability (protected). Updates trek's availableDates.
- **GET /api/bookings:** Returns user's bookings (protected).
- **GET /api/bookings/:id:** Returns booking details.
- **PUT /api/bookings/:id/cancel:** Allows user to cancel booking (with refund logic mock).
- **Admin:** GET /api/bookings/admin to list all bookings, PUT /api/bookings/:id/status to update booking status.

Payment Endpoints:

- **POST /api/payments/initiate:** Initiates payment with gateway, returns payment URL.
- **POST /api/payments/webhook:** Receives payment status updates from gateway, updates booking and payment statuses.

Middleware:

- **authMiddleware:** Verifies JWT and attaches user to request.
- **adminMiddleware:** Checks if user role is admin.
- **validateRequest:** Uses Joi to validate request body/params before reaching controllers.

3.4.4 Frontend Development

I built the customer-facing React application with a focus on user experience and responsiveness.

Component Architecture: I structured components by feature: common/ (Button, Input, Modal), treks/ (TrekCard, TrekList, TrekDetail), booking/ (BookingForm, BookingSummary), user/ (Login, Register, Profile), admin/ (AdminDashboard, ManageTreks, ManageBookings).

State Management: I used Redux Toolkit for global state (user auth, cart-like booking summary). I created slices for auth, treks, bookings, and ui.

Routing: I implemented client-side routing with React Router, including protected routes for authenticated pages and admin-only routes.

Data Fetching: I used RTK Query to define API endpoints and manage caching, loading, and error states. This simplified data fetching and reduced boilerplate.

Trek Listing and Detail: The home page featured a grid of trek cards with images, price, duration. The detail page displayed full itinerary, available dates, and a booking button that opened a modal.

Booking Form: I created a multi-step booking form that collected number of participants, preferred date (calendar picker with only available dates), and special requests. The form validated inputs and showed real-time price calculation.

User Dashboard: After login, users could view their bookings, cancel upcoming trips, and edit profile.

Admin Dashboard: I built a separate dashboard accessible only to admins, with tabs for managing treks, viewing all bookings, and viewing payment reports. I used charts (Recharts) to display booking trends.

3.4.5 Payment Gateway Integration

Integrating a payment gateway was crucial for completing the booking process.

Gateway Selection: I used a mock payment gateway (Stripe test mode) for demonstration. I set up a Stripe account and obtained test API keys.

Backend Integration:

- **POST /api/payments/initiate:** Created a payment intent on Stripe and returned the client secret.
- **POST /api/payments/webhook:** Listened for Stripe webhook events to confirm payment success/failure.

Frontend Integration:

- I used Stripe's React components (@stripe/react-stripe-js) to embed a secure payment form on the checkout page.
- After booking creation, the user was redirected to the payment page where they entered card details (test card 4242 4242 4242 4242).
- On successful payment, the booking status was updated, and the user saw a confirmation page.

Error Handling: I added fallbacks for network issues and payment failures, displaying user-friendly messages and allowing retry.

3.4.6 Admin Dashboard and Content Management

The admin dashboard allowed the client to manage the platform without coding.

Trek Management: Admins could create, edit, and delete treks via forms. Images were uploaded to Cloudinary, and URLs stored in the database.

Booking Management: Admins could view all bookings with filters (date, status, trek). They could update booking status (confirm, cancel) and view payment details.

User Management: Admins could view user list and roles.

Reporting: I added simple analytics showing total bookings, revenue, and popular treks using data aggregated from MongoDB.

3.4.7 Authentication and Security

I implemented robust security measures to protect user data and the platform.

Password Hashing: Used bcrypt with salt rounds to hash passwords before storing.

JWT Tokens: Generated JWT with expiration; stored in HTTP-only cookies (backend) to prevent XSS attacks. I also implemented refresh token rotation for better security.

Input Validation: Used Joi to validate all incoming request data, preventing injection attacks.

Rate Limiting: Added express-rate-limit to limit API requests per IP, preventing brute force attacks.

Helmet: Used Helmet.js to set secure HTTP headers.

CORS: Configured CORS to allow only the frontend domain.

MongoDB Security: Used connection string with credentials and network access rules to restrict access.

3.4.8 Responsive Design and Frontend Polish

Ensuring the website worked seamlessly on all devices was a priority.

Mobile-First CSS: I used CSS Modules and styled components for responsive design. I defined breakpoints in a theme.js file.

Layout Components: I built a ResponsiveGrid component that accepted props for columns per breakpoint.

Navigation: The header used a hamburger menu on mobile; the admin sidebar collapsed on smaller screens.

Booking Flow: The multi-step booking form was designed to be usable on mobile, with large touch targets and clear step indicators.

Testing: I tested on Chrome, Firefox, Safari, and on physical devices (iPhone, iPad, Android). I fixed issues like form fields not zooming on focus, and ensured all interactive elements were accessible.

3.4.9 Performance Optimization

I applied techniques to ensure fast load times and smooth user experience.

Frontend:

- Lazy loaded routes using React.lazy.
- Used react-image with lazy loading for trek images.
- Implemented code splitting for the admin panel (loaded only when needed).
- Optimized bundle by analyzing with webpack-bundle-analyzer and removing unused dependencies.

Backend:

- Added compression middleware to reduce response size.
- Used MongoDB indexing to speed up queries.
- Implemented caching for frequently accessed trek data using Redis (basic setup).

Database: I optimized queries by using projection to return only needed fields and used aggregation pipelines for reports.

3.4.10 Testing and Quality Assurance

I conducted extensive testing to ensure reliability.

Unit Tests: Wrote Jest tests for utility functions and backend models.

Integration Tests: Used Supertest to test API endpoints, covering authentication, booking creation, and payment flows.

Frontend Tests: Used React Testing Library to test critical components (e.g., booking form validation, cart updates).

Manual Testing: I executed a test plan covering all user journeys (browse, book, pay, cancel, admin management). I logged bugs in GitHub Issues and tracked fixes.

Payment Testing: I tested the Stripe integration with test cards (success, failure, 3D Secure) to ensure all scenarios handled gracefully.

3.4.11 Deployment and DevOps

I deployed the application to production and set up CI/CD.

Backend Deployment: I deployed the Node.js API to Heroku, configured environment variables, and set up a MongoDB Atlas cluster.

Frontend Deployment: I deployed the React app to Vercel, connected to the GitHub repository, and configured custom domain.

CI/CD: I set up GitHub Actions to run tests on every push to main and to deploy automatically after passing.

Monitoring: I added Sentry for error tracking and used Heroku logs for backend monitoring.

3.4.12 Documentation and Handover

To ensure the client and future developers could maintain the platform, I prepared comprehensive documentation.

API Documentation: I used Postman to document all endpoints with request/response examples.

Setup Guide: I wrote a detailed README with steps to clone, install dependencies, set up environment variables, and run the project locally.

Admin Manual: I created a user guide for the admin dashboard explaining how to add treks, manage bookings, and view reports.

Training Session: I conducted a live walkthrough with the client to demonstrate all features and answer questions.

3.4.13 Summary of Achievements

During the 10-week internship, I:

- Designed and implemented a full-stack MERN application with over 20 API endpoints.
- Built a responsive frontend with 40+ React components.
- Integrated secure payment processing (Stripe) for online transactions.
- Implemented role-based access control with JWT authentication.
- Achieved a Lighthouse performance score of 90+ for the frontend.
- Wrote comprehensive tests covering critical functionality.
- Deployed the application to production with CI/CD pipelines.
- Delivered a fully functional tourism booking platform that met all client requirements.

These activities provided deep exposure to full stack development, from database design to deployment, and significantly advanced my skills in building scalable, secure web applications.

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES

4.1. Conclusion

The 10-week internship at Sajilo Digital Pvt. Ltd. provided an immersive, comprehensive experience in full stack web development through the creation of the Mount Glacier Trek tourism platform. This project successfully delivered a production-ready application that solves real business problems for a tourism company, demonstrating practical application of the complete MERN technology stack.

The experience of building an application from database design to deployment provided invaluable insights into the complexities and considerations of full stack development. Successfully implementing features like secure payment processing, real-time booking systems, and comprehensive admin interfaces strengthened my understanding of how different system components interact in a production environment.

This internship effectively bridged theoretical knowledge with practical implementation, providing clarity on architectural decisions, security considerations, and performance optimization that are critical in professional software development.

4.2. Learning Outcomes

Technical Skills:

- Full Stack Proficiency: Mastered the complete MERN stack including MongoDB schema design, Express API development, React frontend implementation, and Node.js server management.
- Database Design: Gained expertise in NoSQL database design with MongoDB, including data modeling, indexing, aggregation, and performance optimization.
- API Development: Developed comprehensive RESTful APIs with proper authentication, validation, error handling, and documentation.
- Authentication & Security: Implemented secure JWT-based authentication, password encryption, role-based access control, and secure payment processing.
- State Management: Applied advanced state management patterns using Redux Toolkit for complex application flows.

- Payment Integration: Successfully integrated and tested payment gateway with secure transaction handling.
- Deployment & DevOps: Gained experience in deploying both frontend and backend applications, configuring cloud databases, and implementing basic CI/CD workflows.

Professional Skills:

- Project Architecture: Developed ability to design complete system architecture considering scalability, maintainability, and security.
- Problem-Solving: Enhanced debugging and problem-solving skills across the full technology stack.
- Collaboration: Improved teamwork and communication skills through regular code reviews and collaborative development.
- Documentation: Created comprehensive technical documentation for APIs, database schema, and deployment procedures.
- Client Requirements: Learned to translate business requirements into technical specifications and implementations.
- Time Management: Developed ability to manage complex projects with multiple components and deadlines.

The internship has equipped me with both the technical depth and professional mindset required for a career in full stack web development, providing confidence to tackle complex software projects in professional settings.

REFERENCES

Chodorow, K. (2013). *MongoDB: The definitive guide.* O'Reilly Media.

Hoque, S. (2020*). Full-stack React projects. Packt Publishing.*

Masse, M. (2011). *REST API design rulebook*. O'Reilly Media.

MongoDB Documentation. (n.d.). Retrieved from https://www.mongodb.com/docs/

Node.js Documentation. (n.d.). Retrieved from https://nodejs.org/en/docs/

React Documentation. (n.d.). Retrieved from https://reactjs.org/docs/getting-started.html

Redux Toolkit Documentation. (n.d.). Retrieved from https://redux-toolkit.js.org/

OWASP Foundation. (2023). *OWASP security guidelines*. Retrieved from https://owasp.org

APPENDICES

Screenshots

i. Home page

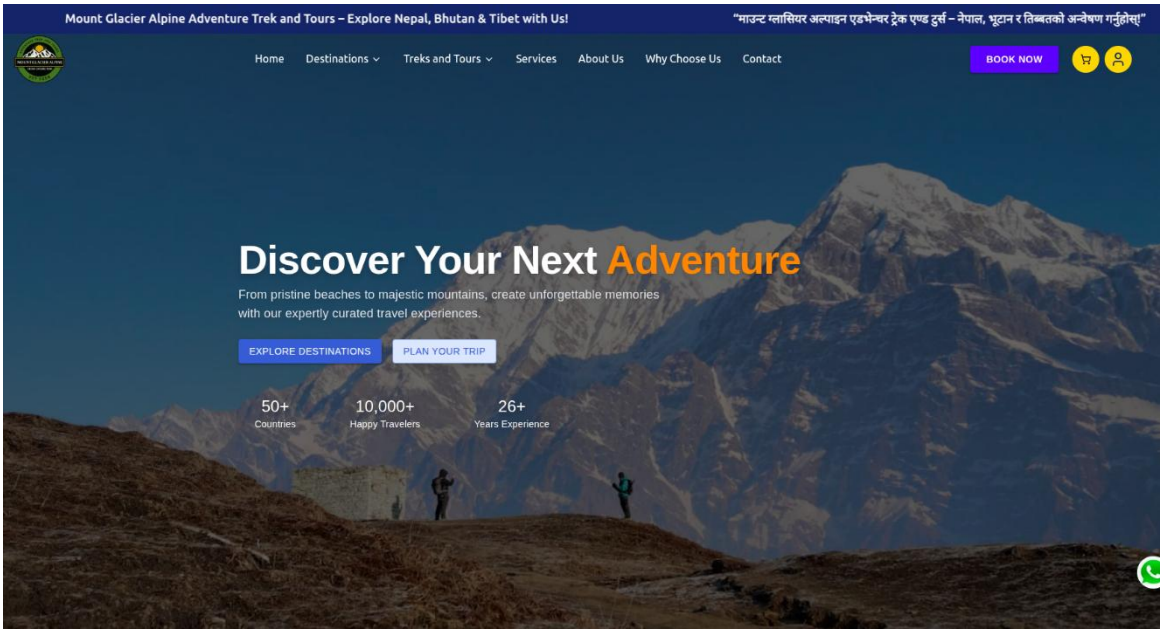

Figure i: Home page of Mount Glacier Trek website

ii. Book adventure

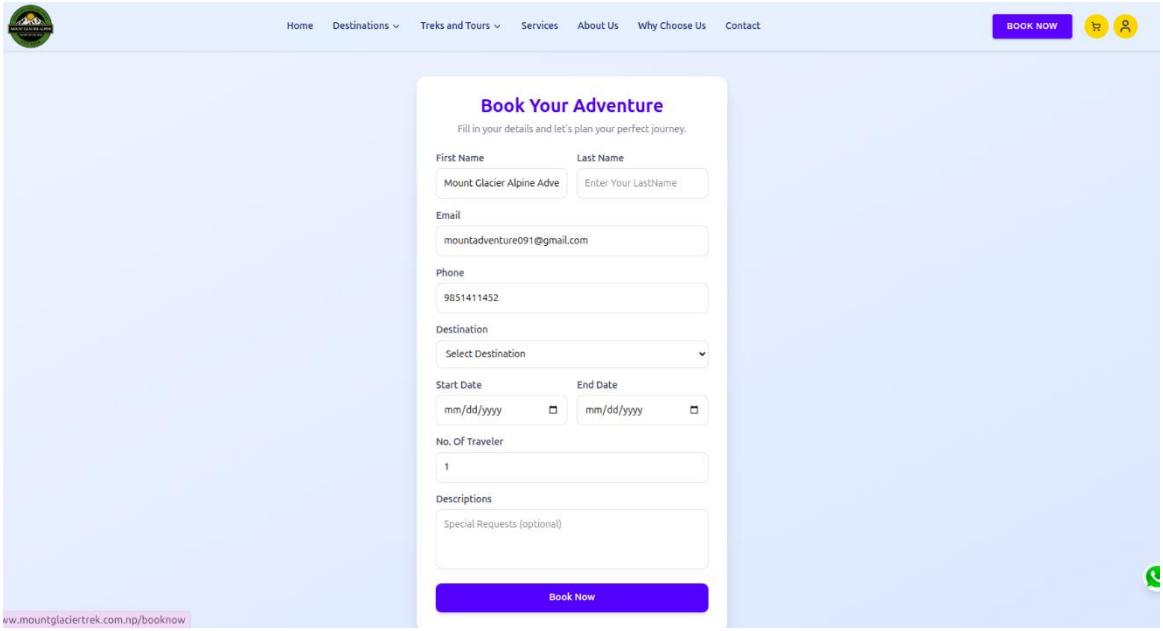

Figure ii: Book Adventure page of Mount Glacier Trek website

iii. Admin Dashboard

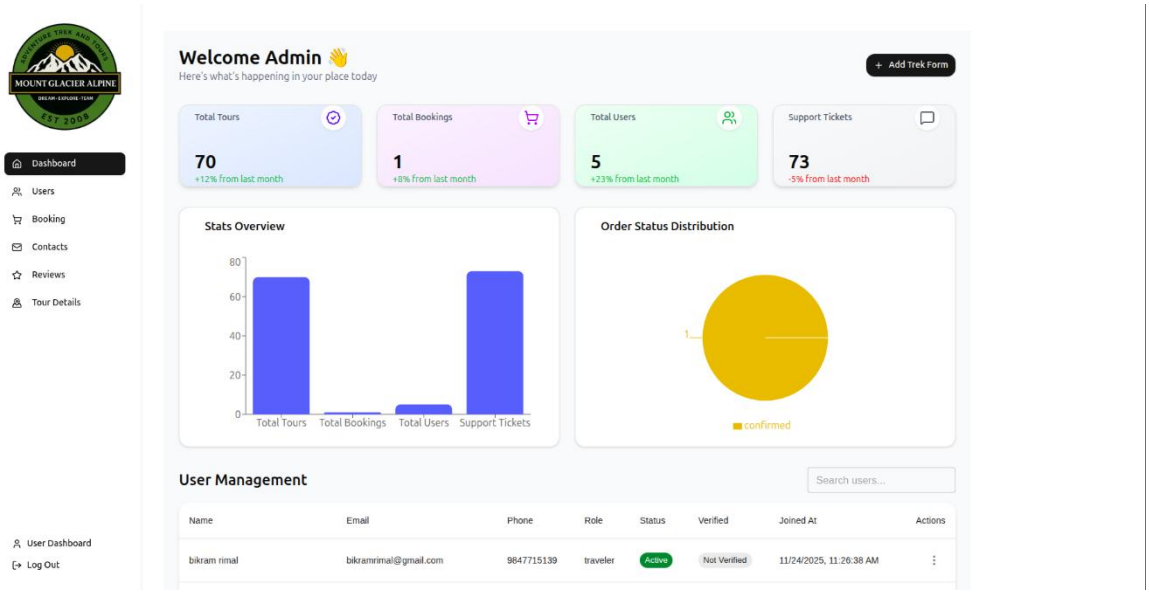

Figure iii: Admin Dashboard of Mount Glacier Trek website

iv. Package page

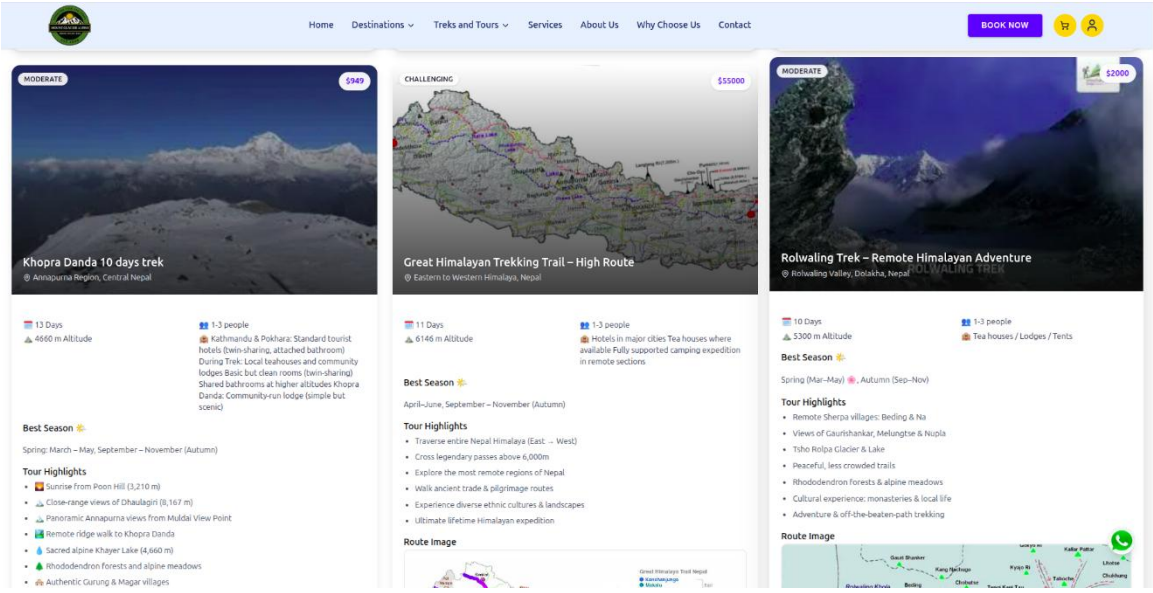

Figure iv: Package page of Mount Glacier Trek website

v. Package Details page

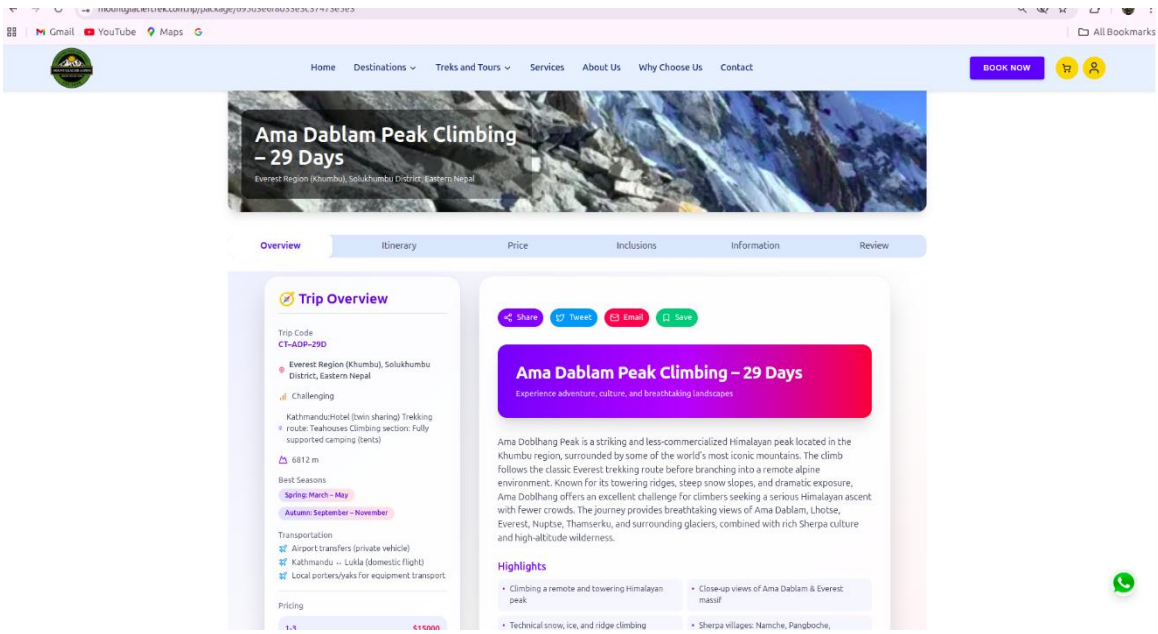

Figure v: Package Detail page of Mount Glacier Trek website

vi. Admin Destination page

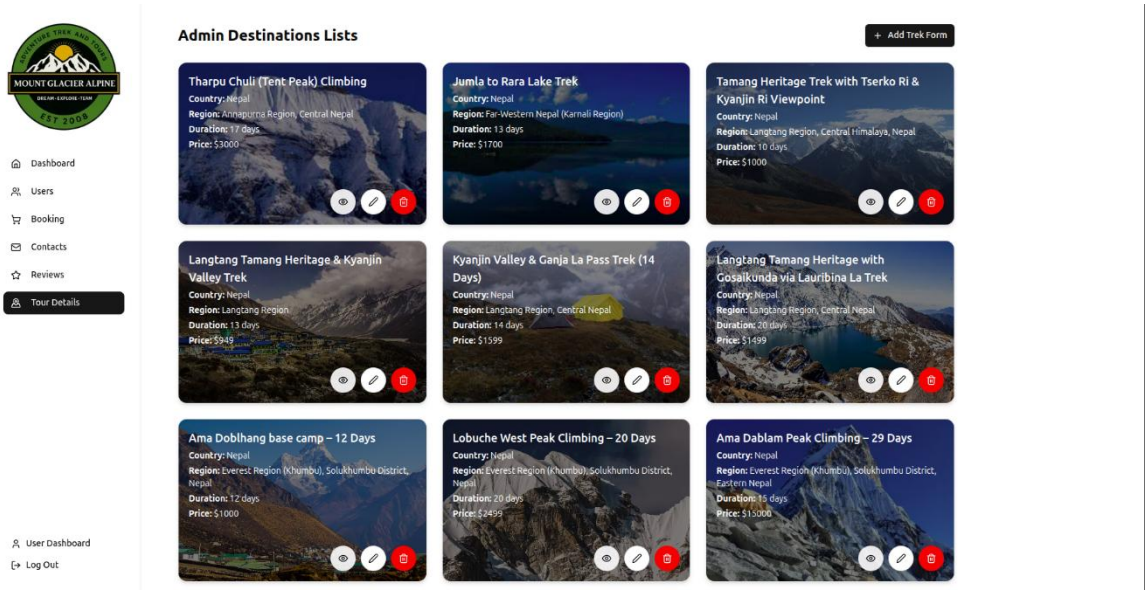

Figure vi: Admin Destination page of Mount Glacier Trek website

vii. Tour List Postmen

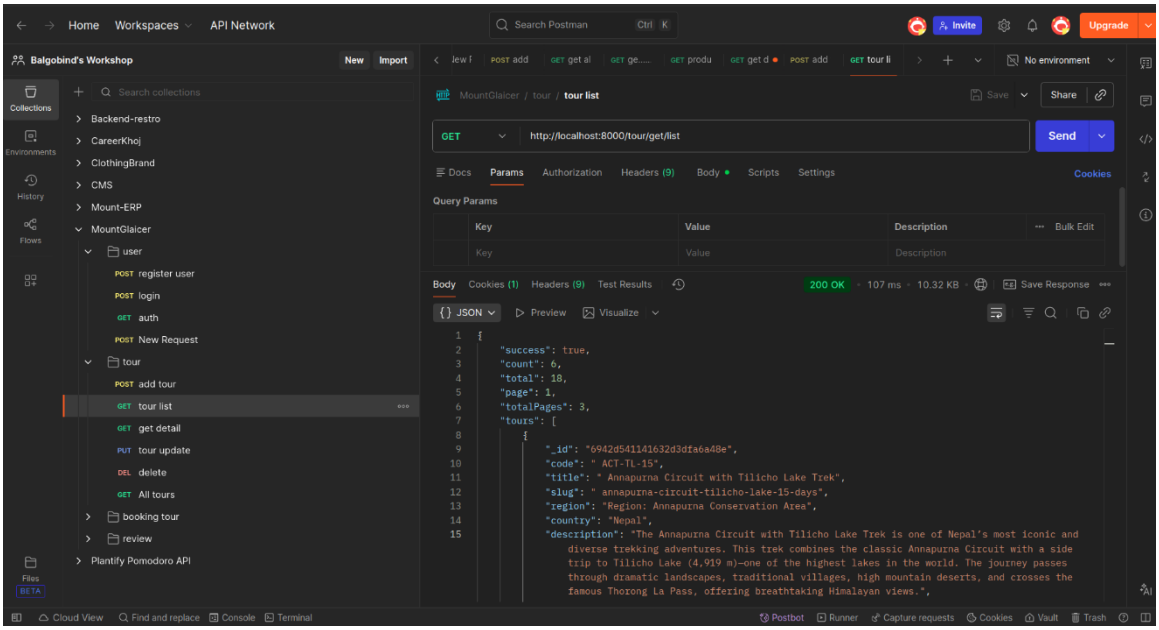

Figure vii: Tour list of Mount Glacier Trek website (Postman)

viii. Add Tour Postman

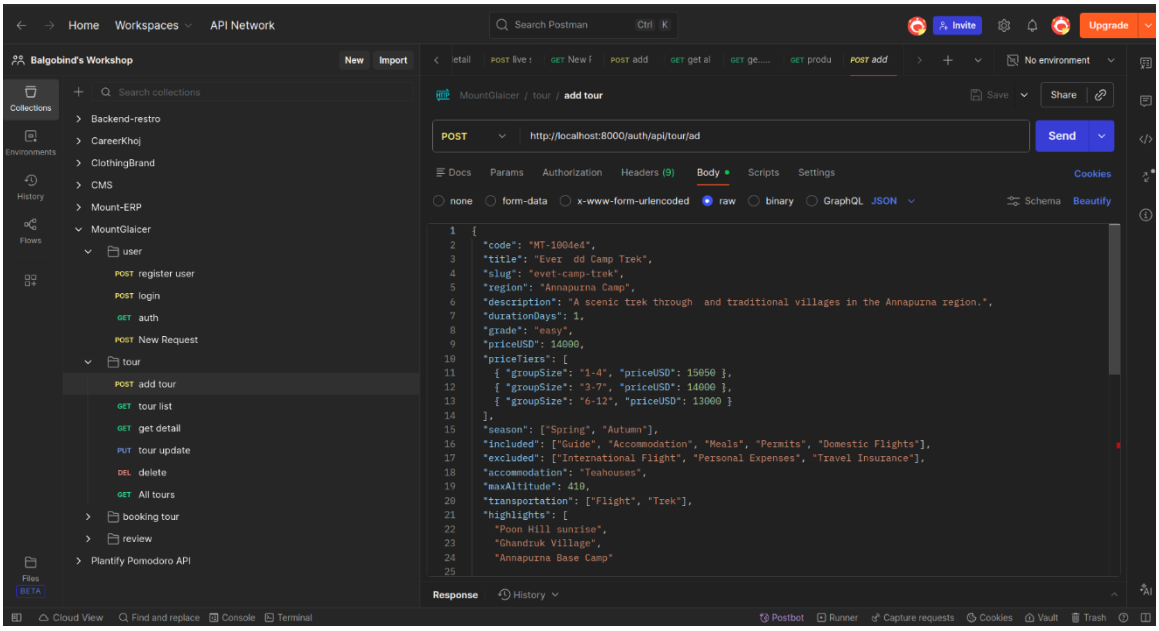

Figure viii: Add Tour of Mount Glacier Trek website (Postman)

ix. Get all Tour Postman

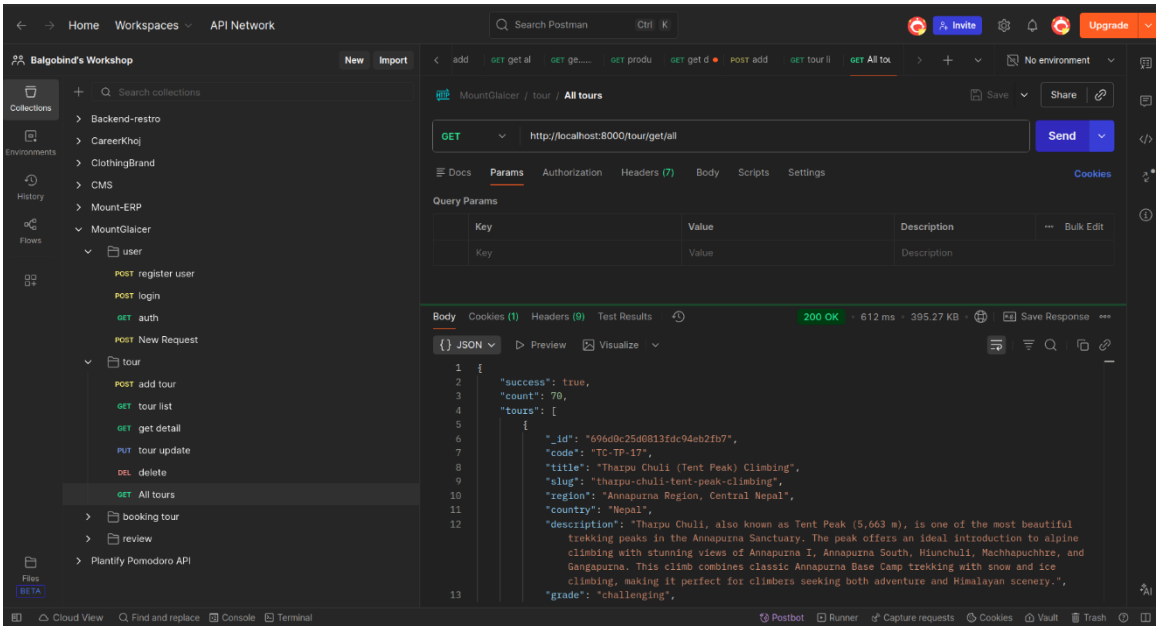

Figure ix: Get all Tour of Mount Glacier Trek website (Postman)